

Football Transfer Market Analysis Using Apache Spark and Machine Learning

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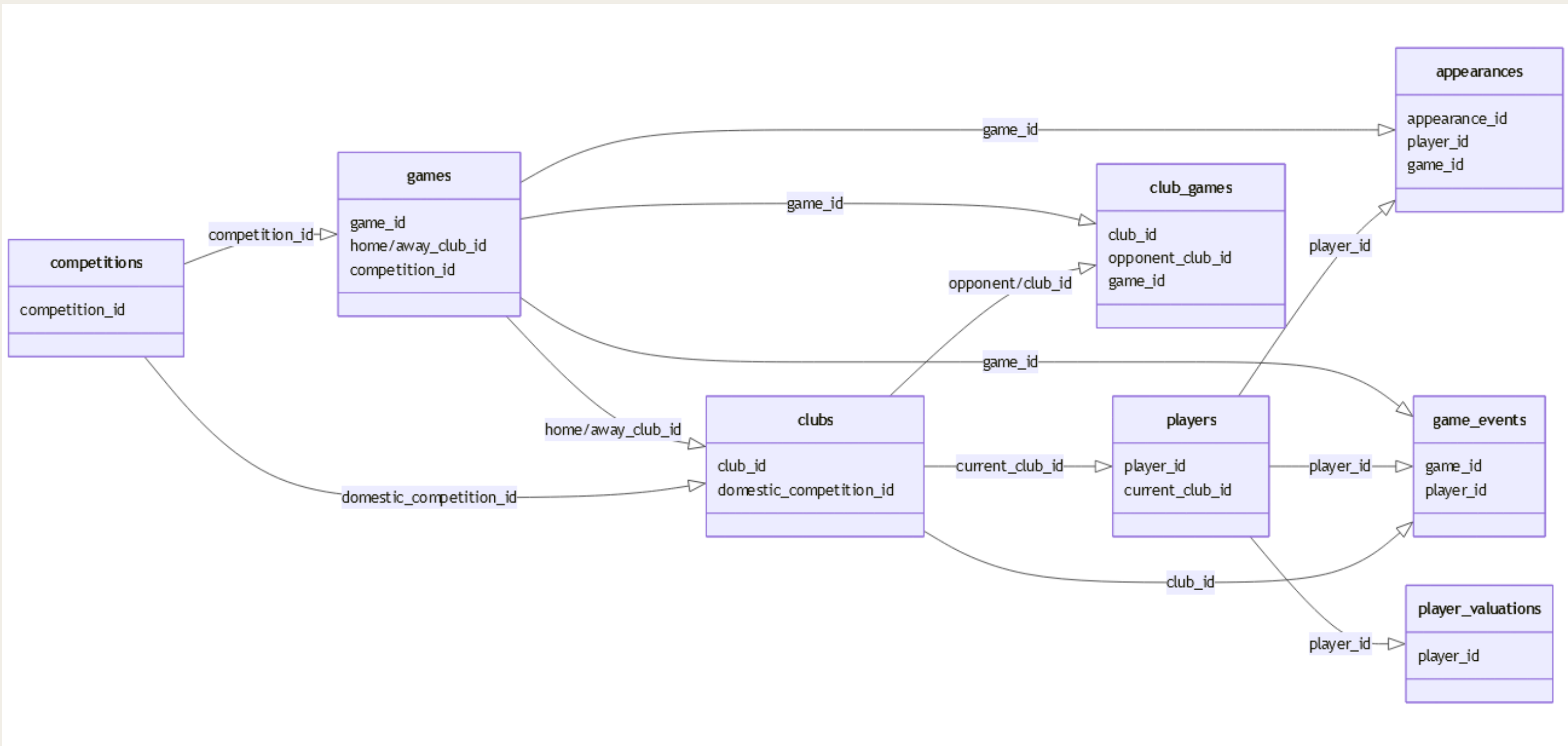
Introduction

The global football transfer market is a dynamic ecosystem driven by player performance, market demands, and financial forces. This project explores the Transfermarkt Player Scores Dataset to analyze historical valuation trends and identify key factors influencing player market value.

I propose to make a predictor model that can make use of qualitative and quantitative data available (of players, clubs, and league) that can predict the transfer fee of any player.

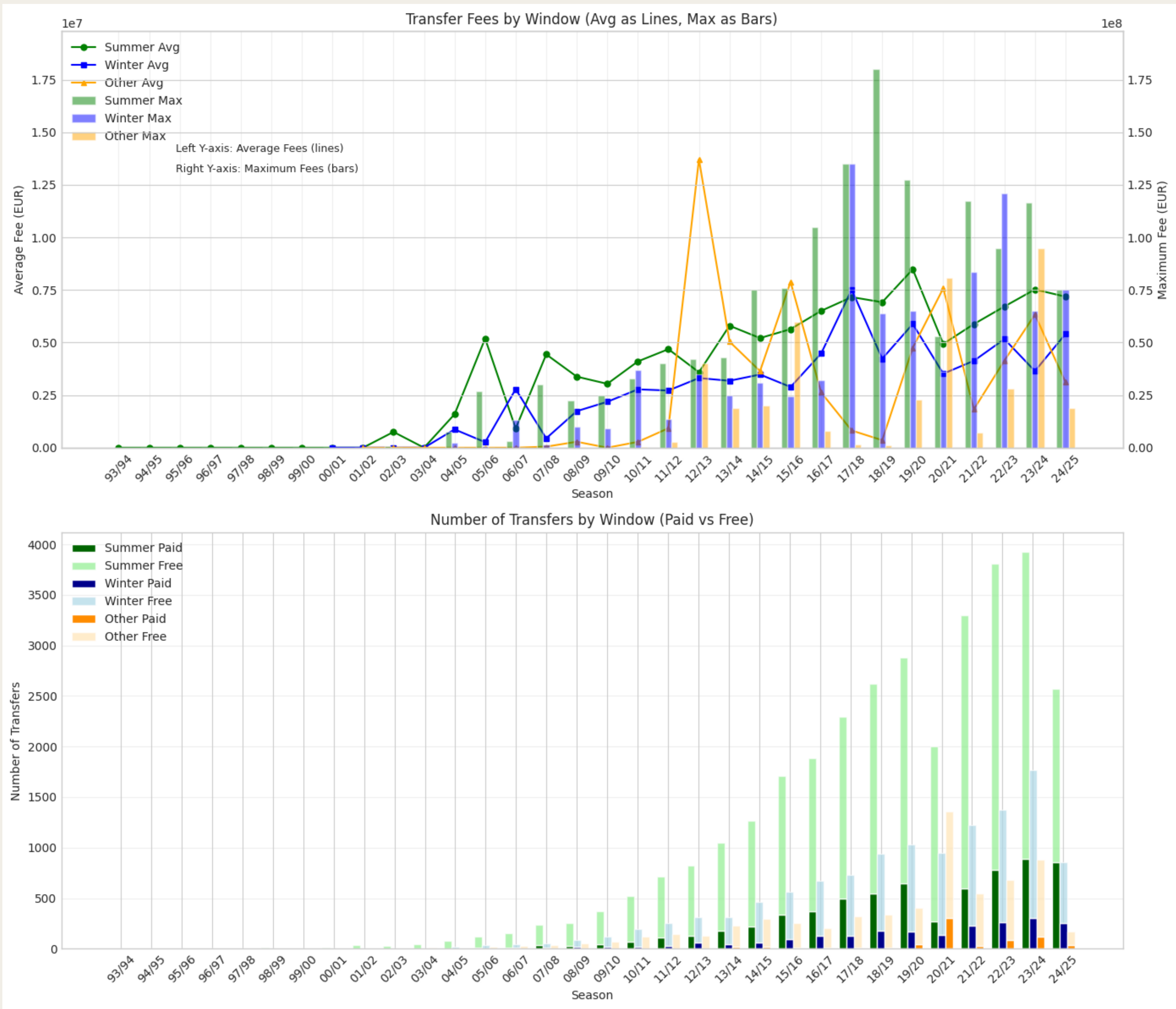
Dataset Overview

- Source:** Kaggle – Player Scores Dataset. Multiple relational CSVs connected via player and club IDs
- Scope:**
 - 60,000+ matches, 400+ clubs, 30,000+ players
 - 400,000 market valuation entries and 1.2 million player appearance records

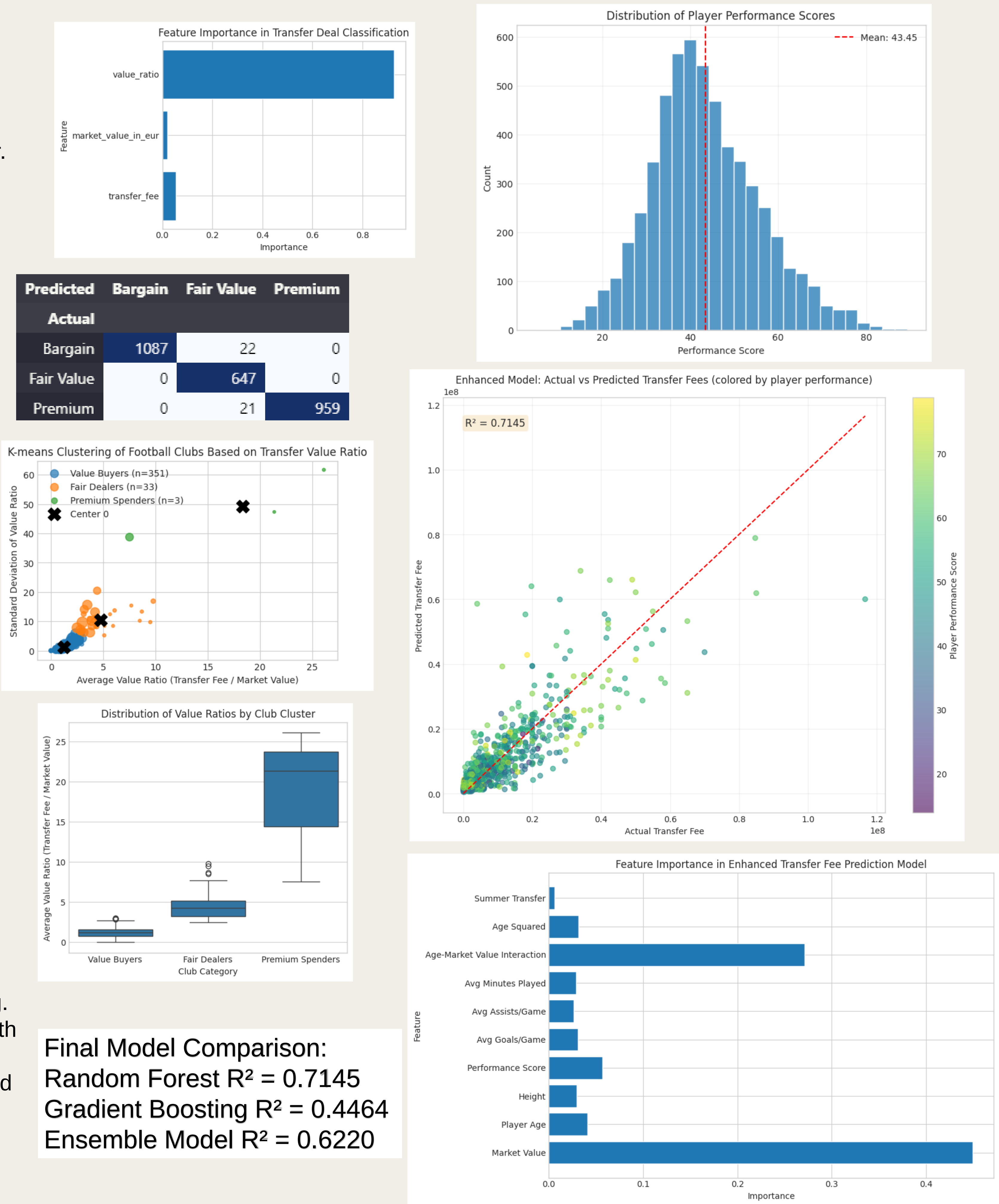


Methods

- Tools Used:** Python with Apache Spark
- Techniques:**
 - Time-series visualization based on league seasons.
 - Financial behavior analysis of clubs to identify value, fair and premium transfers. Evaluated cluster quality and separation.
 - Implemented RandomForest classifier with feature importance analysis
 - Transformed categorical features (position, foot) using indexing and one-hot encoding. Implemented Random Forest Regressor with 100 trees to predict fees for transfers with missing or zero values.
 - Enhanced prediction model incorporating numerical and categorical features. Identified key drivers.
 - Applied Gradient Boosting and Ensemble methods to see if the prediction model performance improves.



Results



Conclusion

- Transfer value is multifactorial, with performance, age, and league context playing critical roles. This study provides a foundation for predictive modeling in sports analytics.
- Summer windows show higher transfer volumes/fees than winter windows. Although, free transfers have in increasing in numbers.
 - Clubs can be classified as value buyers, fair dealers, and premium spenders.
 - Adding player performance metrics improved the prediction model: a 15% increase in R^2 value.
 - With available data metrics, Ensemble model performed better for transfer fee prediction.

Future Work

- Build regression models to predict market value
- Incorporate advanced player performance statistics like expected goals(xG), expected assists (xA), pass per defensive actions (PPDA), etc.
- Explore geographic transfer patterns and agent influence.
- Currently, the player valuations show strong influence on the transfer fee, which is expected. The future works should try to incorporate more player metrics based on match performance, fixture difficulty, etc.

References

1. D. Cariboo, "Football Data from Transfermarkt," Kaggle. [Online]. Available:
2. <https://www.kaggle.com/davidcariboo/player-scores>. [Accessed: May 2, 2025].

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